

3DM Homework Hints and Solutions

Graph Theory Chapter 2

2.2: Hint: Use induction on the length of the closed walk.

2.4: Since $d(u, v)$ is the smallest number of edges on a $u - v$ path and this is always a nonnegative integer, we must have $d(u, v) \geq 0$. If $d(u, v) = 0$ then a shortest $u - v$ path has zero edges, implying that $u = v$. On the other hand, if $u = v$, then the trivial path with zero edges is a shortest $u - v$ path and $d(u, v) = 0$. The last two results collectively give us that $d(u, v) = 0 \iff u = v$. If P is a $u - v$ geodesic, then by following P in reverse we get a $v - u$ path, implying that $d(v, u) \leq d(u, v)$. An identical argument proves that $d(u, v) \leq d(v, u)$; hence, $d(u, v) = d(v, u)$. To see that the last property of a metric space holds, suppose that P is a $u - v$ geodesic and Q a $v - w$ geodesic. By following P from u to v , then Q from v to w we obtain a $u - w$ walk W of length $d(u, v) + d(v, w)$. By Theorem 5, the $u - w$ walk W contains a $u - w$ path P' and thus $d(u, w) \leq \ell(P') \leq d(u, v) + d(v, w)$.

2.5: Suppose, to the contrary, that G is disconnected. Then G contains at least two components. Now ask yourself: What is the smallest possible number of vertices there could be in each component?

2.6: We discussed this question in class.

2.7: To prove that $\text{diam } \overline{G} \leq 2$, let u, v be vertices of G . We need to prove that $d_{\overline{G}}(u, v) \leq 2$. To do this, consider two cases: (1) u and v are in the same component of G , and, (2) u and v are in different components of G . *Now fill in the rest of the proof.*

2.8: We discussed this question in class.

2.10: Hint: From the question, there can't be any pair of vertices at distance 2. What does this tell you about G ?

2.11: To prove that $d(u, v_i) = i$, prove that $d(u, v_i) \leq i$ and then prove that $d(u, v_i) \geq i$.

2.12: Suppose, to the contrary, that there exists a graph G and two longest paths P and Q in G such that P and Q have no vertex in common. Since G is connected, we can find a path R that starts at a vertex u of P , ends at a vertex v of Q , and such that u and v are the only vertices of R that are on P or Q . *Now prove that you can find a path longer than P or Q , a contradiction.*

2.16: The key to this question is the fact that if G is self-complementary, then $G \cong \overline{G}$ and consequently $\text{diam } G = \text{diam } \overline{G}$. It's easy to prove that no nontrivial graph of diameter 1 is self-complementary (*prove it!*), and thus if G is a nontrivial self-complementary graph, then $\text{diam } G \geq 2$. We now prove that it can't be larger than 3. Suppose, to the contrary, that there exists a self-complementary graph G with $\text{diam } G \geq 4$. Then G contains a geodesic P with $\ell(P) = 4$ (*why?*). *Now use this fact to prove that $\overline{G}$ must have diameter less than 4, a contradiction.*

- 2.17: Let u and v be adjacent vertices in a connected graph G . Now use the triangle inequality to prove that if $x \in V(G)$, then $d(u, x) \leq 1 + e(v)$. Then what?
- 2.18: a. This is true. How would you prove it?
b. This is false. What would a counterexample look like?
- 2.19: a. Suppose, to the contrary, that u lies between c and u' . Then $d(c, u') = d(c, u) + d(u, u')$. Why is this a problem? How does this lead to a contradiction?
b. We know that $e(u') \geq e(u)$ (why?). Now consider a $c - u'$ path and think about Question 2.17.